

A. Convert scores to reference quantiles

Observed, non-linked marginals: U (prior) and V (current)

B1 & B2. Estimate subgroup growth regime $\hat{H}_s$ (copula + regime $\Rightarrow$ induced marginal)

Choose $H \in \mathcal{H}_{\text{Beta}}$ to minimize $W_1(F_{V,s}^{\text{obs}}, F_H)$, where $F_H(v) = \mathbb{E}_U [H(F_0(v | U))]$

C. Recover inferred latent SGPC density $P_s \sim \hat{H}_s$

Report density/summary: $\text{meanSGPC} = 100\hat{m}$